

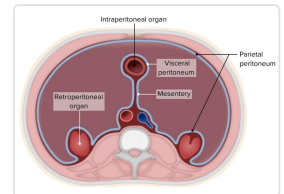
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name the four boundaries of the epiploic foramen.
2. Which nerves supply the parietal peritoneum of the central diaphragm and of the pelvic wall?
3. Where does fluid collect first in a supine patient, and which gutter links it to the pelvis?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Peritoneum — layers and organs

What	Large, thin serous membrane lining the abdominopelvic cavity; built like pleura and serous pericardium
Parietal layer	Lines abdominal and pelvic walls · same vessels as the overlying wall · sensitive to pain, pressure, temperature
Visceral layer	Covers most abdominal organs · sensitive only to stretch and tearing
Peritoneal cavity	Between the layers · ~ 50 mL of lubricating fluid · largest, most complex serous sac
Male vs female	Male: closed sac · female: open to the exterior via uterine tubes, uterus, vagina → pelvic infections more common
Retroperitoneal	On the posterior wall, peritoneum on the front only: kidneys, suprarenals, ureters, pancreas, duodenum, ascending and descending colon
Intraperitoneal	Wrapped all round: stomach, spleen, liver, transverse colon, sigmoid colon, jejunum, ileum



Intraperitoneal and retroperitoneal organs in cross-section.

Peritoneum	Nerve supply
Anterolateral wall	T7-T12 and L1 spinal nerves (same as the overlying skin and muscle)
Diaphragm, central	Phrenic nerve
Diaphragm, peripheral	T7-T12 and L1
Pelvis, peripheral	Obturator nerve
Visceral	Autonomic afferents of the organs it covers

2 Peritoneal folds

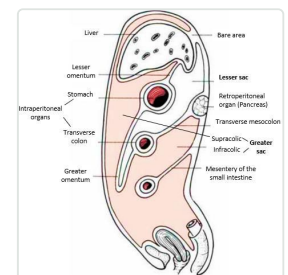
Mesentery / mesocolon	Connects intestine to the posterior abdominal wall
Omentum	Connects the stomach to other organs
Ligament	Condensed fold connecting organs to the wall or to each other

	Greater omentum ("apron")	Lesser omentum
Extent	Greater curvature + 1st inch of duodenum → transverse colon and mesocolon	Lesser curvature, right side of abdominal esophagus, first 2 cm of duodenum → liver (inverted L: fissure for ligamentum venosum + porta hepatis)
Contents	Fat · milky spots (macrophages) · right and left gastroepiploic vessels	Free margin: bile duct (ant-right), hepatic artery proper (ant-left), portal vein (posterior, deepest) · hepatogastric part: gastric vessels, nodes, vagal and celiac fibres

Clinical	"Policeman of the abdomen": walls off infection · fat store · graft material	Free margin = anterior boundary of the epiploic foramen
Fold	Key facts	
Mesentery of small intestine	Fan-shaped · root: DJ flexure (left of L2) → ileocecal junction (right sacroiliac joint) · crosses D3, aorta, IVC, right gonadal vessels, right ureter, right psoas · ~6 m free border pleats around jejunum and ileum · contains SMA, SMV, nodes, fat, nerves · root splits the infracolic compartment into right and left	
Transverse mesocolon	Fused with the back of the greater omentum · divides the cavity into supracolic and infracolic compartments and acts as a barrier to infection · contains middle colic vessels, ascending branches of right and left colic vessels	
Sigmoid mesocolon	Inverted-V root: left limb along the external iliac artery , right limb on the pelvic surface of the sacrum · apex where the left ureter crosses the left common iliac vessels	
Falciform ligament	Sickle-shaped: anterosuperior liver → supraumbilical anterior wall and under-surface of diaphragm · divides the subphrenic space into right and left	

3 Greater sac, lesser sac and the epiploic foramen

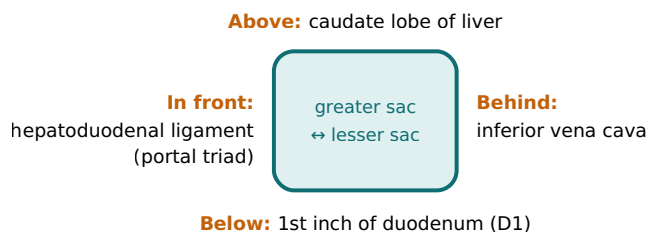
Greater sac	Main compartment · above: diaphragm · below: into the pelvis · front: anterior wall · back: posterior wall (lumbar vertebrae, psoas major, quadratus lumborum)
Lesser sac	Smaller compartment behind the stomach, liver and lesser omentum
Communication	Only through the epiploic (omental) foramen



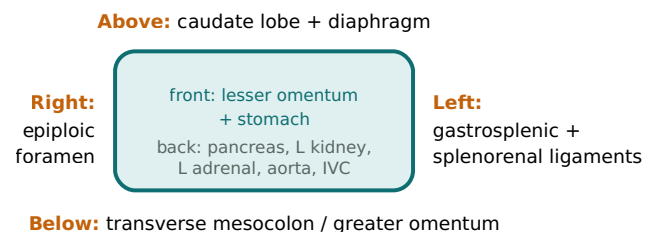
Sagittal section: greater sac, lesser sac and the folds.

BOUNDARY BOXES — the two most-examined peritoneal "rooms"

EPIPLOIC (OMENTAL) FORAMEN



LESSER SAC (OMENTAL BURSA)

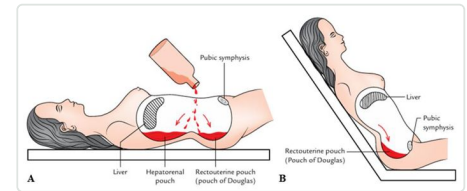


A loop of bowel passing from the greater into the lesser sac herniates through the epiploic foramen.

4 Compartments, spaces and gutters

Why they matter	Collect blood, pus or ascites · routes for spreading infection · sites of internal hernias
Supracolic	Above the transverse colon and mesocolon: liver, stomach, spleen, superior duodenum
Infracolic	Jejunum and ileum, framed by ascending, transverse and descending colon · split into right and left spaces by the root of the mesentery
Paracolic gutters	Right: links the hepatorenal pouch with the pelvis · left: limited above by the phrenicocolic ligament
Subphrenic spaces	Below the diaphragm, above the liver · right and left , divided by the falciform ligament
Duodenal recesses	Superior, inferior, paraduodenal, retroduodenal pockets at the DJ flexure → strangulated internal hernia

Where	Between the posteroinferior surface of the liver and the front of the right kidney
Why it matters	Lowest recess in the supine position (above the pelvic brim) · most spaces drain into it → free fluid, blood, abscess and ascites collect here · pus after upper abdominal surgery (e.g. cholecystectomy) · surgeons place a drain here
Spread	Joined to the rectouterine pouch by the right paracolic gutter → infection passes between abdomen and pelvis



Fluid collects in the hepatorenal pouch when supine.

Hepatorenal pouch = **Morison's pouch** (spelled "Morrison's" on the slides).

P Must-remember pearls

- Parietal peritoneum feels pain, touch, pressure, temperature (somatic) · visceral feels only **stretch and tearing** (autonomic).
- Nerves: anterolateral wall and peripheral diaphragm **T7-L1** · central diaphragm **phrenic** · peripheral pelvis **obturator**.
- Female peritoneal cavity is open via tubes, uterus and vagina → more pelvic infections.
- Retroperitoneal: kidneys, suprarenals, ureters, pancreas, duodenum, ascending and descending colon.
- Lesser omentum free margin: bile duct (ant-right), hepatic artery proper (ant-left), **portal vein (posterior)**.
- Epiploic foramen: **front** hepatoduodenal ligament · **back** IVC · **above** caudate lobe · **below** D1.
- Lesser sac: front lesser omentum + stomach · back pancreas, L kidney, L adrenal, aorta, IVC · left gastrosplenic + splenorenal ligaments.
- Transverse mesocolon divides supracolic from infracolic compartments; mesentery root divides right from left infracolic.
- Sigmoid mesocolon apex = where the **left ureter** crosses the common iliac vessels.
- Hepatorenal (Morison's) pouch = lowest recess supine; right paracolic gutter connects it with the pelvis.
- Subphrenic spaces are divided into right and left by the **falciform ligament**.

Q Test yourself

1. A patient has an internal hernia in which a loop of small intestine passes from the greater sac into the lesser sac. Through which structure did it pass?

- | | |
|-------------------------|-------------------------------|
| A. Transverse mesocolon | B. Epiploic (omental) foramen |
| C. Falciform ligament | D. Hepatogastric ligament |
| E. Sigmoid mesocolon | |

2. After upper abdominal surgery a patient develops an abscess beneath the diaphragm. Imaging shows two compartments separated by the falciform ligament. Which spaces are these?

- | | |
|--------------------------------------|---|
| A. Hepatorenal and subhepatic spaces | B. Rectouterine and vesicouterine pouches |
| C. Right and left subphrenic spaces | D. Supra- and infracolic spaces |
| E. Omental bursa compartments | |

3. A trauma patient lying supine has free intraperitoneal blood. On ultrasound, where is it most likely to be seen first?

- | | |
|----------------------------------|---------------------|
| A. Left paracolic gutter | B. Lesser sac |
| C. Left subphrenic space | D. Infracolic space |
| E. Hepatorenal (Morison's) pouch | |

4. A patient with peritonitis has severe, well-localised pain over the anterolateral abdominal wall. Which nerves carry this pain?

- | | |
|--------------------|-----------------------------------|
| A. Phrenic nerves | B. Autonomic afferents of the gut |
| C. Obturator nerve | D. T7-T12 and L1 spinal nerves |
| E. Vagus nerves | |

5. A surgeon places a finger in the epiploic foramen. What lies directly behind his finger?

- A. Inferior vena cava
- B. Portal vein
- C. Caudate lobe of the liver
- D. First part of the duodenum
- E. Head of the pancreas

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 B — The greater and lesser sacs communicate only through the epiploic foramen, so an internal hernia into the lesser sac passes through it.

2 C — The falciform ligament divides the subphrenic space into right and left subphrenic spaces.

3 E — The hepatorenal pouch is the lowest recess of the abdomen in the supine position, so fluid collects there first.

4 D — The parietal peritoneum of the anterolateral wall is somatic, supplied by T7–T12 and L1, so its pain is sharp and localised.

5 A — The IVC is the posterior boundary; the hepatoduodenal ligament (with the portal vein) is anterior.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy SGL 2 slides, 2026-2027
Study alongside the lecture, not instead of it.